

Human flourishing: an enabler of entrepreneurial intention in Latin American students

Human
flourishing

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Abstract

Purpose – This study aims to propose a causal relationship between the level of students' human flourishing (HF) and their entrepreneurial intention (EI) and the mediation role of entrepreneurial self-efficacy (ESE).

Design/methodology/approach – The study uses a sample of 5,035 first-year university students who graduated from 950 different upper secondary schools in Mexico and abroad. Data were analysed using structural equation modelling.

Findings – The results reveal that HF increases EI and ESE; ESE positively impacts EI; and the student's ESE partially mediates the HF-EI relationship.

Practical implications – These findings may improve educational strategies in upper secondary schools and higher education institutions' management programmes to impact graduates' HF and EI. To the knowledge, the authors are the first to measure how HF in upper secondary school graduates contributes to increasing their EI.

Social implications – Recommendations are made to increase Latin American students' HF and EI.

Originality/value – The authors propose an inverse relationship between HF and EI, as HF can be enhanced in upper secondary education to boost EI later in students' lives.

Keywords Latin America, Entrepreneurial intention, Entrepreneurial self-efficacy, Human flourishing, Upper-secondary education, Self-efficacy, Mexico, Entrepreneurship

Paper type Research paper

1. Introduction

Entrepreneurs are a primary source of economic growth, competitiveness and development (Acs and Amorós, 2008; Romero-Martínez and Milone, 2016). Empirical evidence shows that entrepreneurial activity is associated with higher levels of employment, innovation and productivity (Quatraro and Vivarelli, 2013; Van Praag and Versloot, 2007); therefore, entrepreneurship attracts attention from governments and higher education institutions (HEIs) (Hallam *et al.*, 2016; McNally *et al.*, 2016). The literature on entrepreneurship is vast and diverse, especially with respect to the traits and factors that increase entrepreneurial activity (Audretsch, 2012).

Due to social, environmental and economic conditions in Latin America, entrepreneurship in young people arises from a lack of well-paid job opportunities and as a way to survive (Canales García *et al.*, 2017). It is, therefore, crucial to identify the mechanisms that generate



entrepreneurial intention (EI) with the goal of creating new ventures that can survive and create employment and innovation to benefit society (Guerrero *et al.*, 2018). Furthermore, it is critical to support entrepreneurs in finding opportunities that integrate economic, social and environmental values to create sustainable products and services using inclusive approaches. Education can play a crucial role in this effort.

Entrepreneurship education requires an adequate setting for dedicated entrepreneurship programmes. Adolescence is an ideal stage in human development for acquiring knowledge about creating ventures and developing a favourable attitude towards entrepreneurship (Peterman and Kennedy, 2003). Thus, entrepreneurship education in upper secondary education can influence entrepreneurship and increase EI rates in students (Garrido-Yserte *et al.*, 2020). Research has shown a positive reaction from students at the upper secondary level towards increasing entrepreneurial competencies, intentions, career choices, abilities and entrepreneurial self-efficacy (ESE) (Aamir *et al.*, 2019). For this study, it is relevant to highlight that upper secondary and undergraduate education may play an essential role in students' personal growth, ESE and EI.

In this regard, human flourishing (HF) as a determinant of EI may be a way to promote entrepreneurship in Latin America. Potential or active entrepreneurs who manifest higher HF levels increase their entrepreneurial passion, and therefore, have a greater chance of accomplishing their business goals. These high HF entrepreneurs may create ventures with a better chance of surviving, growing and benefitting society. Therefore, upper secondary schools, HEIs (Oades *et al.*, 2011) and business programmes should train students to include HF in their academic curricula as a critical element of entrepreneurship education (McKenna and Biloslavo, 2011).

To our knowledge, few studies have examined the relationships amongst EI, ESE and HF in the Latin American context. Therefore, this research aims to test whether HF has a positive effect on EI and if ESE mediates the relationship between HF and EI. We conducted an online survey of 5,035 first-year students at HEIs who graduated from 950 different upper secondary schools in Mexico and abroad. Structural equation modelling (SEM) was used to analyse data.

This study makes two main contributions. Firstly, according to Wiklund *et al.* (2019), the study of HF in the context of entrepreneurship is still emergent. In our work, HF is analysed as a cause of EI, not a consequence as it has traditionally been investigated (VanderWeele *et al.*, 2020), providing a broader understanding of the topic. Secondly, although a substantial number of studies have been conducted on entrepreneurship education, the literature on this topic still lacks content at the primary and upper secondary levels (Aamir *et al.*, 2019). Our findings suggest that HF increases perceived EI and that ESE partially mediates this relationship.

The remainder of this study is structured as follows: Section 2 provides a review of the literature and our hypotheses regarding their relationships. Section 3 describes the methodology and data sample. Finally, our results, implications, conclusions and future lines of research are presented in Sections 4 and 5.

2. Literature review

2.1 Entrepreneurial intention

EI is the desire to create a new venture, either for economic, business, family, social or personal reasons. It tends to decrease with income (Saiz-Álvarez and Rodríguez-Aceves, 2019) and is a prerequisite to entrepreneurial behaviour according to Ajzen's parsimonious theory of planned behaviour (TPB) (Wathanakom *et al.*, 2020).

Understanding what drives individuals to pursue entrepreneurial careers and engage in entrepreneurial behaviour has been a significant inquiry within the field of entrepreneurship, especially within the educational setting. One way to approach this question is from the

perspective of cognitive theories that capture the way individuals think (Ajzen, 1991). This approach is based on two assumptions:

- (1) Planned behaviour is intentional.
- (2) Intention is the best predictor of behaviour because intentions focus the individual's attention on a given behaviour (Krueger and Brazeal, 1994).

A vast body of research shows that the decision to become an entrepreneur is voluntary and conscious (Krueger *et al.*, 2000; Liñán *et al.*, 2011); that is, new ventures are not created by accident; they result from specific, intentional decisions (Bird, 1988; De Clercq *et al.*, 2011).

Both empirically and conceptually, intentions have been the best predictor of any planned behaviour (Ajzen, 1991; Bagozzi and Yi, 1988; Heuer and Liñán, 2013; Krueger, 1993; Krueger, 2000; Krueger and Brazeal, 1994; Krueger and Carsrud, 1993; Krueger *et al.*, 2000; Liñán and Chen, 2009; Liñán *et al.*, 2013; Liñán and Fayolle, 2015; Murugesan and Dominic, 2014; Zampetakis and Moustakis, 2006). EI reflects the desire to act and a belief in behavioural execution (Krueger and Carsrud, 1993). In other words, intentions both precede and predict behaviour, intervening as a mediating variable for action (Krueger and Carsrud, 1993). Additionally, intentions represent the degree to which an individual commits to a future behaviour (Armitage and Conner, 2001; Gelderen *et al.*, 2008; Krueger, 1993) and the effort he or she is willing to expend on it (Ajzen, 1991). Furthermore, because they are based on perceptions, they are learned, which implies that they vary between individuals and situations (Krueger *et al.*, 2000).

TPB is widely used in the field of entrepreneurship as it has been shown to have relevant predictive power for behaviour (Krueger and Carsrud, 1993; Liñán and Fayolle, 2015). According to the TPB, three independent factors determine behavioural intentions:

- (1) Attitude towards the behaviour.
- (2) Social norms.
- (3) Perceived control over the behaviour or self-efficacy (Ajzen, 1991; Krueger and Brazeal, 1994; Krueger *et al.*, 2000; Kolvereid and Isaksen, 2006; Gelderen *et al.*, 2008; Guerrero *et al.*, 2008).

As a general rule, the stronger the intention to carry out behaviour, the greater the likelihood of developing effective action (Ajzen, 1991). Thus, the TPB affirms that a strong attitude, subjective norms and perceived control over a given behaviour strengthen the intention to carry out a particular action (Autio *et al.*, 2001; Díaz-García and Jiménez-Moreno, 2010; Kolvereid, 1996). Because of its predictive capacity with respect to EI, we analyse ESE in the next subsection.

2.2 Entrepreneurial self-efficacy

ESE is a belief in one's own ability to perform tasks and assume roles associated with entrepreneurial outcomes (Chen *et al.*, 1998; Gist, 1987). In other words, it refers to one's perception of one's capacity to carry out a job or a series of tasks (Krueger, 2000). Thus, ESE influences behavioural expectations and aspirations to fulfil personal goals (Krueger and Carsrud, 1993). It is a motivational construct that influences the individual's choice of activities and goals, as well as persistence and performance in various contexts (Zhao *et al.*, 2005). Therefore, it is a cognitive variable that estimates how entrepreneurs can be motivated, with the help of cognitive resources, to take actions to control life events (Chen *et al.*, 1998).

From a domain-specific perspective, ESE is appropriate for the study of the entrepreneurial phenomenon as it has great explanatory power in predicting the intentional

strength to start a business and the probability that EI will be transformed into entrepreneurial actions (Boyd and Vozikis, 1994; Chen *et al.*, 1998; Mueller and Dato-On, 2008). Consequently, ESE has emerged as a critical psychological construct in entrepreneurship research (Newman *et al.*, 2019), as it influences entrepreneurial motivation, intention, behaviour and performance. In addition, it is a fundamental outcome measure of entrepreneurship training and education. Given that ESE can be influenced by education and training, there is evidence of the construct's malleable nature.

Newman *et al.* (2019) identify two types of ESE antecedents – at the individual level (work experience, education and training, role models, mentors and individual differences) and the firm and macro-level (firm's characteristics and the cultural and institutional environment). Furthermore, they also identify individual-level ESE outcomes (EI, entrepreneurial emotions/mental state and entrepreneurial actions/behaviours) and firm-level outcomes (entrepreneurial performance and venture creation). Whilst entrepreneurial emotions and mental state have been studied as self-efficacy outcomes, they have been approached from the perspective of entrepreneurial passion (Cardon and Kirk, 2015) and by framing uncertainty as an opportunity (Engel *et al.*, 2014).

To our knowledge, ESE has not been linked to HF, either as a possible consequence or an antecedent. Additionally, we found no study that considered ESE in a mediating role between HF and EI. ESE plays an essential role in our research as it is crucial in determining whether an individual will pursue an entrepreneurial career and engage in entrepreneurial behaviour (Newman *et al.*, 2019).

2.3 Human flourishing

HF is a recurring theme in civilisation's history and has been approached from different disciplines, including economics, political philosophy, moral philosophy and psychology. In ancient Greece, the philosopher Aristotle referred to *eudaimonia*, a term that has been translated as flourishing, happiness or well-being (Charles-Leija *et al.*, 2018, 2020; Henrekson, 2014; Jayawickreme *et al.*, 2012; Ryff, 2019; Wiklund *et al.*, 2019). In this context, HF refers to individuals' ability to reach their full potential in life. In other words, flourishing means the achievement of goods, purposes and ends of human existence (Messer, 2013) or as a state in which most aspects of a person's life are good (VanderWeele, 2017; VanderWeele *et al.*, 2020). The act of flourishing has been described as "living within an optimal range of human functioning, one that connotes goodness, generativity, growth and resilience" (Fredrickson and Losada, 2005, p. 1).

Recent studies have associated the concept of HF with aspects of human functioning, such as personal relationships (Deci and Ryan, 2000; Diener *et al.*, 2010; Huppert and So, 2013; Ryan and Deci, 2000; Ryff, 2019; Seligman, 2012), purpose and meaning (Diener *et al.*, 2010; Huppert and So, 2013; Ryff, 2019; Seligman, 2012), environmental mastery (Diener *et al.*, 2010; Huppert and So, 2013; Ryan and Deci, 2000; Ryff, 2019) and happiness and life satisfaction (Vuckovic, Floyd and Riley, 2019), as well as experiencing positive emotions, such as optimism and vitality (Diener *et al.*, 2010; Huppert and So, 2013; Seligman, 2012). These have been included in previous studies as indicators that individuals are reaching their full potential and leading a good life; that is, they are experiencing HF.

It is essential to notice that several conceptualisations and HF measures have been developed by previous scholars (Keyes, 2002; Diener *et al.*, 2010; Seligman, 2012; Huppert and So, 2013; Hone *et al.*, 2014; VanderWeele, 2017). In this study, we use Diener *et al.*'s (2010) Flourishing Scale for its conciseness (eight items) and because several studies have indicated it is a valid and reliable instrument to measure HF. These validations have been made in Spanish (Checa *et al.*, 2018), French (Villieux *et al.*, 2016), Italian (Giuntoli *et al.*, 2017), Chinese (Duan and Xie, 2019),

Portuguese (Silva and Caetano, 2013) and Japanese (Sumi, 2014). The Flourishing Scale has also been used in various contexts such as clinical (Schotanus-Dijkstra *et al.*, 2016), general population (Checa *et al.*, 2018), adolescents (Duan and Xie, 2019; Silva and Caetano, 2013), college students (Sumi, 2014) and employees inside organisations (Imran *et al.*, 2020). Additionally, the scale has been validated in nationally representative surveys (Hone *et al.*, 2014). Other studies have found that HF increases employees' work engagement and resilience (Imran *et al.*, 2020), decreases turnover and increases productivity (VanderWeele *et al.*, 2020). In the educational setting, HF has been studied as an outcome (Ouweneel *et al.*, 2011) operationalised by student engagement. Moreover, previous research proposes that HF is the ideal educational aim (Kristjansson, 2017). In this study, we propose HF as a predictor of EI, mediated by ESE. In the following section, we discuss the relationships between HF, EI and ESE.

2.4 The link between human flourishing and entrepreneurial intention

Individuals endowed with higher HF levels generally feel a sense of purpose and meaning in their lives, are optimistic and resilient, have good personal relationships and a propensity to view events positively that may influence their self-motivation to achieve goodness. Conversely, entrepreneurship is the process of identifying opportunities in the marketplace, marshalling resources to pursue those opportunities and committing to take actions and using resources to exploit the opportunities for long-term personal gain (Parker, 2004; Gartner, 1989). It is also defined as a human-creative act that builds something of value from practically nothing, which requires vision, commitment and specific competencies and abilities (Boyd and Vozikis, 1994). Thus, individuals with high HF levels have a greater chance of identifying opportunities (entrepreneurial or otherwise) and taking advantage of them (Lyubomirsky *et al.*, 2005). They can acquire skills and abilities to improve their performance (as the leader of a new venture or in other areas) because they have meaning and purpose. Moreover, they remain optimistic in the face of adversity and setbacks, which is essential given that starting a venture is challenging. Finally, individuals with higher HF levels usually dedicate more time to personal and family relationships, thereby increasing their network, which is generally used to acquire the resources a new venture needs to survive and grow (McLure-Wasko and Faraj, 2005). Thus, we hypothesize the following:

H1. HF positively impacts EI.

2.5 The link between human flourishing and entrepreneurial self-efficacy

A variety of HF elements may contribute to ESE. Firstly, having strong relationships with family and friends contributes to one's perception of having sufficient economic and emotional resources to start a venture (Davidsson and Honig, 2002; Marshall *et al.*, 2020). Secondly, being respected and trustworthy increases the possibility of building new enriching relationships with key people who may fund a venture (Lewicki and Bunker, 1995; Moysidou and Hausberg, 2020). Thirdly, when people use their main strengths and abilities, they tend to be interested and engaged in their daily activities (Gute *et al.*, 2008), which improves their ability to identify business opportunities and innovative ideas. Fourthly, having a purpose and meaning in life encourages one's determination to reach personal, professional or entrepreneurial goals (Hill *et al.*, 2016). Finally, people with HF are competent and capable when engaged in activities that allow them to interact effectively with their environment (White, 1959); in other words, they have environmental mastery (Ryff, 1989). Having environmental mastery implies that "one can manage the surrounding environment, including making effective use of available opportunities, whilst also creating contexts

suitable to one’s personal needs and values” (Ryff, 2019, p. 649). Based on this, we hypothesize the following:

H2. HF positively impacts ESE.

2.6 The link between entrepreneurial self-efficacy and entrepreneurial intention

ESE influences one’s expectations regarding the result of engaging in certain behaviours and one’s aspirations for obtaining that result (Krueger and Carsrud, 1993). It is a motivational construct that influences the choice of an individual’s activities, objectives, persistence and level of performance in various contexts (Zhao et al., 2005). In this regard, ESE has strong explanatory power for entrepreneurial behaviour, as it can predict both the strength of EIs and the probability that these intentions will be realised (Boyd and Vozikis, 1994; Chen et al., 1998; Mueller and Dato-On, 2008). In fact, when entrepreneurs are confident that they can perform the tasks necessary to create a new venture, they are more likely to successfully perform those tasks (Cardon and Kirk, 2015). In line with previous studies (Hsu et al., 2019; Tsai et al., 2016), we hypothesize the following:

H3. ESE positively impacts EI.

2.7 The mediating role of entrepreneurial self-efficacy

Although ESE has not been studied as a mediator of HF and EI, research shows that when individuals flourish, they are mindful of their strengths, build better and stronger relationships (Myers, 2000), are more resilient (Seligman, 2012) and are more capable of identifying opportunities (Fredrickson, 2009), as alertness is crucial in doing so (Tang et al., 2012). Moreover, the mediating role of ESE has been studied in terms of the relationships between various constructs and EI and has always shown strong results, including EI and social media (Huang and Zhang, 2020); perceived barriers (Dolarслан et al., 2020); entrepreneurial mentoring (Cho et al., 2020); personality traits (Kusumawijaya, 2018); narcissism, psychopathy and Machiavellianism (Wu et al., 2019); information technologies convergence education (Jung, 2018); innovativeness (Priyankara et al., 2017) and university educational support and mentoring (Zhicheng et al., 2017). In sum, ESE, HF and EI are closely related. Therefore, we hypothesize the following:

H4. ESE mediates the relationship between HF and EI.

Our hypotheses are explained graphically in Figure 1.

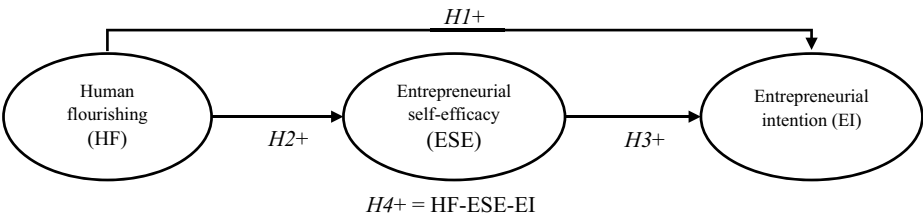


Figure 1.
Research model

Source: Authors

3. Methodology

Our research design comprises an empirical study in which quantitative data were collected via an online survey adapted from previously tested instruments. Data were analysed using SEM. Details of the sample and data collection are described as follows.

3.1 Sample and data collection

Survey respondents were freshmen students at a private university in Mexico who graduated from 950 different upper secondary education institutions located throughout the country and abroad. The surveys were conducted online using Qualtrics and 5,035 observations were collected in Week 6 of the 2020 fall semester (which runs for 18 weeks from August to December). Respondents were enrolled at 26 different campuses across the country; 35% live in the country's centre, 23.3% in the West, 22.4% in the Northeast, 12.8% in the Northwest and 6.5% live abroad. The respondents' average age was 18.3 years; 55.9% were men and 44.1% were women.

The questionnaire used a Likert scale with anchor points of 1 = "total disagreement" and 7 = "total agreement". On average, respondents required 15 min to complete the survey. The following subsection describes the instrument used to measure the concepts of interest.

3.2 Measurement and constructs

Previously validated scales were adapted and used to measure the components of the research model. The survey items and their sources are listed in [Appendix](#). The dependent variable EI comprises four items built from a scale adapted from [Mueller \(2011\)](#) and [Liñán and Chen \(2009\)](#). HF, the independent variable, is measured with items adapted from the scale in [Diener et al. \(2010\)](#). The mediating variable ESE is measured using five items developed by [Mueller \(2011\)](#). Finally, gender (female = 1, male = 0) is included as a control variable. The following paragraphs describe the process of analysing the data and obtaining our results.

3.3 Analysis and results

The means, standard deviations (SD) and correlations between the constructs and the control variable are presented in [Table 1](#). Data imputed from the confirmatory factor analysis (CFA) model demonstrated in [Table 2](#) was used for such an aim. To impute data, we follow the procedure in [Arbuckle \(2010\)](#). Because of the use of Likert scales to measure the main constructs, their means and SD are relatively similar.

All correlations amongst the main constructs are below 0.7 and significant ($p < 0.01$). The highest is between ESE and HF, at 0.638. Gender is significantly correlated with EI, HF and ESE; surprisingly, it is positively correlated with HF but negatively correlated with EI and ESE, suggesting that female respondents are more likely to have a higher HF ($r = 0.061$). Variance inflation factors were calculated to test for multi-collinearity. The results ranged from 1.00 to 1.710 and as all values are below 10, as [Asteriou and Hall \(2011\)](#) suggested, multi-collinearity is not a concern.

3.4 Exploratory factor analysis, confirmatory factor analysis and structural equation modelling

The research model was created using statistical package for social sciences 22 and analysis of a moment structures ([Arbuckle, 2010](#)) with a maximum likelihood estimation. Firstly, the measuring scales' psychometric properties (reliability, convergent and discriminant validity) were evaluated using a CFA. Next, the structural model was assessed to test $H1$ –

H4, Hand multi-group MSEM was used to test the control variable (Anderson and Gerbing, 1988). To assess reliability, internal consistencies (Cronbach’s α) for the variable’s scales were computed. To create the CFA’s best possible measurement model, it was necessary to drop two items, namely, HF2 and HF8 (Appendix). The remaining items and their standardised loadings are presented in Table 2.

Almost all items have large and significant factor loadings. The composite reliability scores range from 0.87 to 0.89 and the average variance extracted values range from 0.52 to 0.66. The shared variance is less than the variance extracted by the indicators that measure the three constructs (Fornell and Larcker, 1981; Hair *et al.*, 2010) and Cronbach’s alphas are greater than 0.7, the threshold suggested in Bagozzi and Yi (1988). In sum, reliability, convergent and discriminant validity are confirmed (Table 2).

The goodness of fit indices are all within the recommended thresholds (Cheung and Rensvold, 2002; Hair *et al.*, 1986). The normed χ^2 , comparative fit index, goodness of fit index, incremental fit index and root mean square error of approximation are 18.1, 0.968, 0.960, 0.968 and 0.058, respectively, providing support for the overall goodness of fit indices (Table 2). Following Podsakoff *et al.* (2003), we apply Harman’s one-factor test to test for

Table 1.
Means, standard
deviations and
correlations

#	Variables	Mean	SD	1	2	3	4
1	Entrepreneurial intention (EI)	4.72	1.57	1			
2	Human flourishing (HF)	6.48	0.806	0.398***	1		
3	Entrepreneurial self-efficacy (ESE)	5.31	0.976	0.634***	0.638***	1	
4	Gender	0.44	0.497	−0.034**	0.061***	−0.047***	1

Notes: $N = 5,035$; ***, ** and * indicate significance at 1, 5 and 10%, respectively

Table 2.
Exploratory and
confirmatory factor
analysis

Variables	Items	Standardised loading	Squared multiple correlations	α	CR	AVE	Goodness of fit
Entrepreneurial intention (EI)	EI1	0.87	0.76	0.88	0.89	0.66	CFI = 0.968
	EI2	0.71	0.50				IFI = 0.968
	EI3	0.93	0.87				RMSEA = 0.058
	EI4	0.72	0.52				GFI = 0.960
Human flourishing (HF)	HF1	0.74	0.55	0.87	0.87	0.52	Normed $\chi^2 =$ 18.1
	HF3	0.75	0.56				
	HF4	0.69	0.47				
	HF5	0.71	0.50				
	HF6	0.73	0.53				
	HF7	0.73	0.53				
Entrepreneurial self-efficacy (ESE)	ESE1	0.82	0.67	0.90	0.89	0.63	
	ESE2	0.83	0.68				
	ESE3	0.78	0.61				
	ESE4	0.77	0.60				
	ESE5	0.77	0.59				

Notes: α is Cronbach’s α ; CR = composite reliability; AVE = average variance extracted. Seven-point Likert scales were used (1 = strongly disagree; 7 = strongly agree) for all measurement items

common method bias, which compares the standardised regression weight estimates from the CFA model with and without a common latent factor. The single-factor model's fit was entirely unacceptable; therefore, we conclude that method effects do not affect the results in this study.

Figure 2 demonstrates the results of the SEM that tests the relationships amongst the variables simultaneously. The model is acceptable, as shown by the normed $\chi^2 = 18.1$, CFI = 0.968, IFI = 0.968, GFI = 0.960 and RMSEA = 0.058. Of the three paths, two were significant at the 1% level ($H2$ and $H3$) and one at the 5% level ($H1$).

As demonstrated in Figure 2, the results support $H1$ – $H3$. Hand partially supports $H4$. For $H1$, the relationship between HF and EI is significant, but with a small effect ($\beta = 0.038$, $p < 0.01$). For $H2$, HF exerts a positive and significant effect on ESE ($\beta = 0.573$, $p < 0.01$). Regarding $H3$, ESE explains 33.9% of EI's variance with a positive and significant effect ($\beta = 0.560$, $p < 0.01$). As to $H4$, ESE partially mediates the relationship between HF and EI, i.e. HF better explains EI through ESE.

3.5 Results of the control variable

To test the stability of the hypothesized HF–ESE–EI relationship by gender, we use a multi-group analysis to test for differences in the relationships for women (2,222) and men (2,813) separately. The χ^2 of each constrained model was significantly higher than in the unconstrained model, indicating a difference in the HF–ESE–EI relationship across the two groups. Specifically, the path with the most significant difference is from HF to ESE. Results are reported in Table 3.

The following section discusses these results.

4. Discussion

This study explores whether HF positively impacts EI ($H1$) and ESE ($H2$), if ESE positively impacts EI ($H3$) and the mediation role of ESE between HF and EI ($H4$) in a sample of upper secondary graduates in Mexico. The results demonstrate that HF positively impacts EI and ESE. Therefore, $H1$ and $H2$ are validated. As ESE positively impacts EI, $H3$ is also supported. These results are in line with the fact that individuals endowed with higher levels of HF usually have a sense of purpose and meaning, an optimistic view, resilience, good personal relationships and a mood propensity towards positive events that may influence their self-motivation to achieve goodness (Diener *et al.*, 2010; Ryff, 2019). These characteristics are essential for increasing ESE, which is the belief in one's ability to

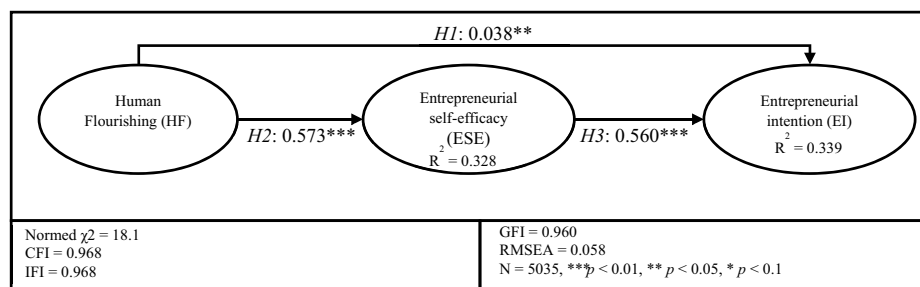


Figure 2.
SEM results

Notes: $N = 5,035$; ***, ** and * indicate significance at 1%, 5% and 10%, respectively

Moderator variable	Description	χ^2	df	$\Delta\chi^2$	Δdf	P-value
Gender	Unconstrained model	1,637.02	168			
	Constrained model A: The path of HF to ESE is specified as equal across groups	1,642.67	169	5.65	1	0.017**
	Constrained model B: The path of ESE to EI is specified as equal across groups	1,638.07	169	1.05	1	0.307 n.s
	Constrained model C: The path of HF to EI is specified as equal across groups	1,637.39	169	0.37	1	0.547 n.s
	Constrained model D: The above paths are fixed as equal across groups	1,644.33	171	7.31	3	0.063*

Table 3.
Control variable
multi-group analysis

Notes: The total sample size is 5,035 with 2,222 women and 2,813 men; $\Delta\chi^2$: difference in χ^2 value between unconstrained and constrained models; Δdf : difference in the number of degrees of freedom; n.s.: non-significant; ***, ** and * indicate significance at 1, 5 and 10%, respectively

successfully pursue venture creation activities based on a personal assessment of one's entrepreneurial skills (Chen *et al.*, 1998).

In addition, ESE mediates the relationship between HF and EI (*H4* is partially validated). It has been found that when individuals flourish, they are mindful of their strengths, build better and stronger relationships and are more resilient and more capable of identifying opportunities (Myers, 2000; Seligman, 2012; Fredrickson, 2009). Furthermore, studies reveal that when students are confident in their abilities to create a new business, they are more likely to succeed in that effort (Cardon and Kirk, 2015). Therefore, an individual endowed with a higher level of HF has a greater perceived ESE, and thus, a greater EI because of their awareness of their capabilities (Ouweneel *et al.*, 2011).

Regarding the control variable, the effect of HF on ESE is statistically more significant in men than in women. A growing number of investigations reveal that women tend to possess lower ESE than men (Chen *et al.*, 1998; Kirkwood, 2009; Wilson *et al.*, 2009) and this gender gap emerges in young adults and adolescents (Kickul *et al.*, 2008; Wilson *et al.*, 2007). Whilst people with high HF are optimistic about the future, research shows that female university students tend to express less passion or reduced physiological arousal for founding a business (Bandura, 1986) compared with their male counterparts (Renko, 2010). Although women receive positive verbal reinforcement (Bandura, 1986) regarding their entrepreneurial capabilities from their supportive and rewarding social relationships (Diener *et al.*, 2010), entrepreneurship tends to be portrayed as a masculine undertaking (Calas *et al.*, 2009; Greene *et al.*, 2001), increasing the ESE gender gap (Dempsey and Jennings, 2014).

Hence, although correlations show that women in the sample have a higher HF than men, its effect on ESE is smaller.

4.1 Methodological implications

The methodology used here highlights the integration of three instruments of reliability and validity, namely, Diner *et al.* (2010), Liñán and Chen (2009) and Mueller (2011). To our knowledge, this is the first work that measures a manifestation of HF in students as it contributes to increasing EI. We note that the scale has been used on multiple occasions for validation processes; however, its use to measure HF as a determinant of other variables of interest is still rare. Our study contributes to filling this gap.

4.2 Theoretical implications

One of the theoretical implications of this study arises from verifying HF's multiple impacts on concepts relevant to the literature on entrepreneurship and business. Most studies on HF have focussed on its conceptualisation, measurement and determinants. Our research contributes to understanding HF as a cause of favourable elements for entrepreneurship and society in general. Given the growing interest in well-being within firms (Bellet *et al.*, 2019), this article contributes and helps to position HF as a relevant topic for business studies.

4.3 Practical implications

At a practical level, our study uses an educational context to focus on HF as a contributor to the development of new ventures through its impact on ESE and EI. When an institution contributes to students' HF, it increases their academic success (Adler, 2017), strengthens their EI, generates economic growth and provides employment.

Our findings have other important practical implications, especially in Latin America, as individuals with higher HF levels are confident professionals who care not only about themselves but also for their peers and their social environment. Consequently, if they become entrepreneurs, they are likely to be aware of and involved in sustainable entrepreneurship initiatives that generate economic, social and environmental value, which is urgently needed, especially in emerging economies. Entrepreneurial activities can contribute to solving complex social and ecological problems, and sustainability-oriented entrepreneurs act as catalysts for transformation (DiVito and Ingen-Housz, 2019).

Another important implication is that students' profiles and previous training can make a difference in programmes focussed on fostering EI. Therefore, it is essential to understand such differences and make the necessary modifications in entrepreneurship programmes taught in HEIs. Entrepreneurial education focussed on HF should be applied at all educational levels (Adler, 2017; McKenna and Biloslavo, 2011; Seligman and Adler, 2018). Doing so will increase ESE, fostering new venture creation and contributing to economic growth and social welfare. One helpful strategy could be "positive education" to identify students' character strengths, make them more aware of their abilities and encourage use of those abilities (Peterson and Seligman, 2004). We also note that the frequent use of personal skills contributes to students' well-being and professional success (Harzer and Ruch, 2014).

4.4 Limitations and future research

Our study is limited to Mexican HEIs, and therefore, lacks comparisons with other Latin American countries. Future research efforts will focus on two areas. The first will differentiate between ex-ante and ex-post HF. Here, we dealt with ex-ante HF only, namely, how potential entrepreneurs from upper secondary schools are encouraged to pursue

entrepreneurship, with a perspective based on generating a positive impact due to HF. Future research will focus on how HF is defined by entrepreneurs from different cultures to compare HF and EI in developed and developing countries.

The second line of research would propose a new definition related to HF and entrepreneurship. This definition will promote perspectives on how a constructive mental disposition and positive attitude towards starting new ventures are related to social relationships, self-efficacy, purpose in life, commitment and optimism. Other studies could replicate the findings in this study in other countries to bolster their relevance to a broader audience.

5. Conclusions

This study aimed to show that HF is a predictor of EI and that ESE mediates that relationship. ESE has strong explanatory power with respect to entrepreneurial behaviour (Boyd and Vozikis, 1994; Chen *et al.*, 1998; Mueller and Dato-On, 2008) but has not been linked to HF as a consequence or an antecedent. Although HF has been principally studied as an outcome (VanderWeele *et al.*, 2020), some research has studied HF as a predictor, especially in the organisational setting. Complementary to that latter body of research, we propose an inverse relationship between HF and EI, as HF can be enhanced in upper secondary education to boost EI later in students' lives. We have found that HF increases EI and ESE; ESE positively impacts EI; and students' ESE partially mediates the HF–EI relationship. Given that ESE can be influenced by education and training, there is evidence of its malleable nature. HF may promote sustainable entrepreneurship in Latin America by acting as a determinant of EI, opening new research lines to find alternatives linked to HF.

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<i>Student's entrepreneurial intention (EI)</i>	
<i>(Mueller, 2011; Liñán and Chen, 2009)</i>	
EI1	I plan to start a new business within five years of completing my studies
EI2	I have already made/taken some steps towards starting my own business (e.g. seeking information, discussing the idea with friends and writing a business plan)
EI3	I am sure I will start my own business within five years of completing my studies
EI4	It is one of my career goals to become an entrepreneur
<i>Student's human flourishing (HF)</i>	
<i>(Diener et al., 2010)</i>	
HF1	I lead a purposeful and meaningful life
HF2	My social relationships are supportive and rewarding
HF3	I am engaged and interested in my daily activities
HF4	I actively contribute to the happiness and well-being of others
HF5	I am competent and capable in the activities that are important to me
HF6	I am a good person and live a good life
HF7	I am optimistic about my future
HF8	People respect me
<i>Student's entrepreneurial self-efficacy (ESE)</i>	
<i>(Mueller, 2011)</i>	
ESE1	I feel I am able to define an idea and a business strategy for a new venture
ESE2	I feel I am able to write a business plan (conduct a market research study, a financial analysis, etc.)
ESE3	I feel I am able to negotiate and maintain supportive relationships with potential investors and banks
ESE3	I feel I am able to recognise opportunities for the development of new products and/or services
ESE4	I feel I am able to build relationships with key people to obtain the capital needed to start a new business

Table A1.
Measurement scales
used

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